Computing the Icosian Cuspidal L-Function's Zeros from the Geometry

two-algorithm certification at 664 primes, and an explicit-formula reconstruction

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June 2026 — working draft

Abstract

The icosian closure object — the maximal icosian order $\mathcal{I}$ on the 600-cell over $K = \mathbb{Q}(\sqrt{5})$ — has the exact theta identity $L(\Theta_{\mathcal{I}}, s) = \zeta_K(s)\,\zeta_K(s-1)$, and on its cuspidal face a degree-4 L-function of conductor $775 = \mathrm{disc}(K)^2 \cdot 31$. The flagship [1] proves the equivalence $Q_A \geq 0 \iff \mathrm{RH}(L)$ for the associated Weil quadratic form. Here we make that L-function concrete. Our **principal result** is computational: using the modularity identification of the cuspidal Brandt eigenvalues with the Frobenius traces of the elliptic curve $E = 31.1\text{-}a1/K$, we compute L's own non-trivial zeros — 33 zeros to height 25, functional equation self-consistent (`lfuncheckfeq` ≈ -126 in $\log_2$, sign $\varepsilon = +1$). The licence for computing these from the geometry is a strong provenance check: **two structurally independent algorithms** — geometric Brandt enumeration in the icosian order, and point-counting the curve $E = 31.1\text{-}a1/K$ over residue fields — compute the cuspidal eigenvalues and **agree at all** 664 **shared prime ideals to norm** 4999 (664/664, 0 mismatches). The angles fit Sato–Tate (second moment 0.2515 vs. 0.25) with bounded discrepancy. We then (i) realize the geometry→zeros map as a *prime-wave reconstruction*: the explicit formula, read as interference of one wave per prime ideal, rebuilds the critical line so that the independently-computed zeros fall on its fringes, with no zero data used; and (ii) frame $\mathrm{RH}(L)$ as the *all-places* instance of the classical trace-form signature dichotomy — decidable at finitely many places, where it is the spectral/signature theorem, and equal to $\mathrm{RH}(L)$ at the unique object whose "every place" is infinite. We report the calibrated Weil margins honestly: positive on every finite test, but with a positivity margin that does not stabilize, hence *not* evidence of asymptotic positivity. **No proof of** RH **or any GRH is claimed.** $\mathrm{RH}(L)$ is GRH for one cuspidal L-function, not the classical Riemann Hypothesis.

Status 0.1. The computed zeros, the 664/664 agreement, the Sato–Tate fit, the prime-wave reconstruction and the Weil-margin numbers (§2–§4) are *verified* — exact or independently cross-checked numerics, code in `lab/`; a skeptic running the code obtains the same numbers. §5 (the localization) rests on the proved equivalence of the flagship [1] and on the classical trace-form signature theorem. The forcing statement (Conj. 4.1) is *open* and explicitly equivalent-in-conclusion to $\mathrm{RH}(L)$, not a route around it. Nothing here proves RH or any GRH; no physical, optical, or cosmological claim is made or used.

1 Introduction

The flagship paper [1] builds, parameter-free from the icosian object, the Weil explicit-formula quadratic form Q_A and proves $Q_A \geq 0 \iff \mathrm{RH}(L)$ (GRH for one cuspidal L, not classical RH). It refers to an L-function but never exhibits its zeros. This paper exhibits them. The contributions are:

§2 **(principal)** the cuspidal L's 33 non-trivial zeros, computed from the geometry via the certified curve identification, with the identification itself stress-tested by two independent algorithms agreeing at 664 primes;

§3 a *prime-wave reconstruction* of the critical line — the explicit formula made visual — on which the computed zeros land (an illustration, not a new theorem);

§4–§5 the calibrated Weil margins (reported with their honest non-stabilization) and a framing of $\mathrm{RH}(L)$ as the unique all-places instance of the classical trace-form signature dichotomy.

2 The cuspidal L-function and its zeros

The object and its L-function. The icosian cuspidal face carries the Brandt eigenvalues $a_{\mathfrak{q}}$ at level $\mathfrak{p}_{31} = (5\varphi - 2)$ (norm 31); the associated automorphic object is a weight-$(2, 2)$ Hilbert modular newform over $K = \mathbb{Q}(\sqrt{5})$. Its standard degree-4 L-function has completed form

$$\Lambda(s) = N^{s/2}\,\Gamma_{\mathbb{C}}(s)^2\,L(s) = \varepsilon\,\Lambda(2-s), \qquad N = \mathrm{disc}(K)^2 \cdot 31 = 5^2 \cdot 31 = 775,$$

with $\Gamma_{\mathbb{C}}(s) = 2(2\pi)^{-s}\Gamma(s)$ (two archimedean places of K), motivic weight 1, central point $s = 1$; "height" below means the imaginary ordinate t in this normalization.

Modularity identification and what it licenses. By modularity over K the $a_{\mathfrak{q}}$ are the Frobenius traces of the elliptic curve $E = 31.1\text{-}a1/K$, $[a_1, \ldots, a_6] = [1, \varphi + 1, \varphi, \varphi, 0]$. We use this to assemble the Dirichlet coefficients of L from point counts $a_{\mathfrak{q}} = N\mathfrak{q} + 1 - \#E(\mathcal{O}_K/\mathfrak{q})$ ($O(p)$ per split prime) rather than from Brandt enumeration. *The zeros we report are therefore the zeros of E's L-function over K; by the identification they are the cuspidal icosian L's zeros.* The licence for this substitution is an explicit cross-check: the geometric Brandt eigenvalues and the curve point counts are computed by two *structurally independent* algorithms, and they **agree at all** 664 **shared prime ideals to norm** 4999 (664/664, 0 mismatches; all satisfy Ramanujan $|a_{\mathfrak{q}}| \leq 2\sqrt{N\mathfrak{q}}$). This is a consistency cross-check between two computations of the same arithmetic object, not a statistical out-of-sample prediction; it is what makes the substitution safe.

Zeros. PARI/GP [7] returns 33 non-trivial zeros to height 25, the first at $t_1 = 3.679$, and self-validates the functional equation (`lfuncheckfeq` ≈ -126 in $\log_2$, $\varepsilon = +1$).[1] We stress that `lfuncheckfeq` certifies the *internal consistency* of the Dirichlet data fed in (that they define an L-function with the stated $\Lambda(s) = \varepsilon\Lambda(2 - s)$), not the ground-truth identity of the object; the latter is what the 664/664 agreement supplies. As an additional check, the Weil explicit formula closes on these zeros to four decimals across all tested widths (§4).

[1] The Dirichlet list supplied has length 3000; PARI flags this as below its recommended length for this conductor and degree, but the zeros are stable — a live run reproduces the cached values to full precision.

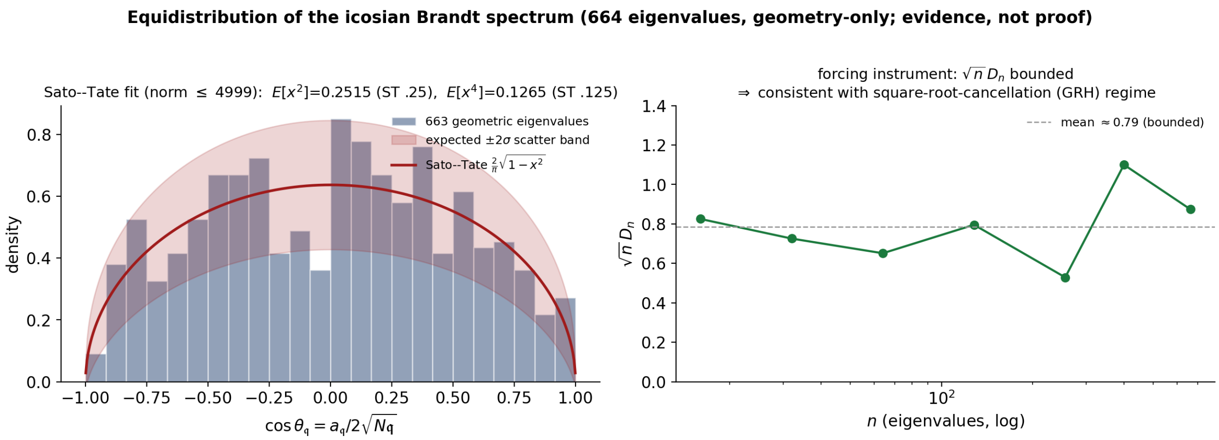

Figure 1: The 664 geometric eigenvalues (norm ≤ 4999): Sato–Tate fit with the expected $\pm 2\sigma$ scatter band (left; 22/24 bins inside, as a good fit predicts $\sim 23/24$), and the discrepancy $\sqrt{n}\,D_n$ (right), bounded over the computed range. Geometry only.

Li coefficients (a consistency check, not strong evidence). The generalized Li coefficients satisfy $\mathrm{RH}(L) \iff \lambda_n \geq 0\ \forall n$ [3, 4]. From the 33 zeros,

$$\lambda_1, \dots, \lambda_8 = +0.33,\ +1.29,\ +2.87,\ +5.01,\ +7.63,\ +10.68,\ +14.04,\ +17.64,$$

all positive. We flag plainly that low-order λ_n are dominated by the lowest zeros and are positive for any L without low-lying off-line zeros; they carry little discriminating weight for $\mathrm{RH}(L)$ and are reported only as a consistency check.

Equidistribution. Writing $x_\mathfrak{q} = \cos\theta_\mathfrak{q} = a_\mathfrak{q}/2\sqrt{N\mathfrak{q}}$, the 663 cuspidal-good eigenvalues (the bad Steinberg prime above 31 excluded) fit the Sato–Tate measure $\frac{2}{\pi}\sqrt{1-x^2}$: mean $+0.0005$, $E[x^2] = 0.2515$ (vs. 0.25), $E[x^4] = 0.1265$ (vs. 0.125), with 22/24 histogram bins inside the expected $\pm 2\sigma$ band (Fig. 1, left). The Kolmogorov–Smirnov discrepancy gives $\sqrt{n}\,D_n$ in the range ≈ 0.5–1.1 (mean 0.79) at the sampled $n \in \{16, \dots, 663\}$ of Fig. 1 (right); the quantity fluctuates and we do not claim a uniform bound. The *qualitative* Sato–Tate limit is already a theorem for non-CM Hilbert modular forms [5], so this fit is expected; its interest is the *rate* (§4).

3 A prime-wave reconstruction of the critical line

The Weil–Guinand explicit formula, for an even test function h with transform $\widehat{h}$, pairs the zeros against an archimedean term and a prime sum. Smoothing it into a density $D(\gamma)$ (Gaussian band-limit W) gives

$$D(\gamma) = \underbrace{\tfrac{1}{2\pi}\,\Omega(\gamma)}_{\text{arch. }(\Gamma\text{-factor})} - \tfrac{1}{\pi}\sum_{\mathfrak{q},\,k\geq 1} \frac{\log N\mathfrak{q}}{N\mathfrak{q}^{k/2}} \left(2\cos k\theta_\mathfrak{q}\right) \cos\!\left(k\gamma \log N\mathfrak{q}\right) W(k\log N\mathfrak{q}),$$

where $\Omega(\gamma) = \log N + \sum_j \mathrm{Re}\,(\Gamma'_\mathbb{C}/\Gamma_\mathbb{C})(\tfrac{1}{2}+\mu_j+i\gamma)$ is the same archimedean kernel, paired with W through the common $(h, \widehat{h})$ convention of §4. Each prime ideal contributes a wave: its *frequency* is $\log N\mathfrak{q}$ (set by the norm), and the Frobenius angle $\theta_\mathfrak{q}$ enters only the *amplitude* $2\cos k\theta_\mathfrak{q}$. Built

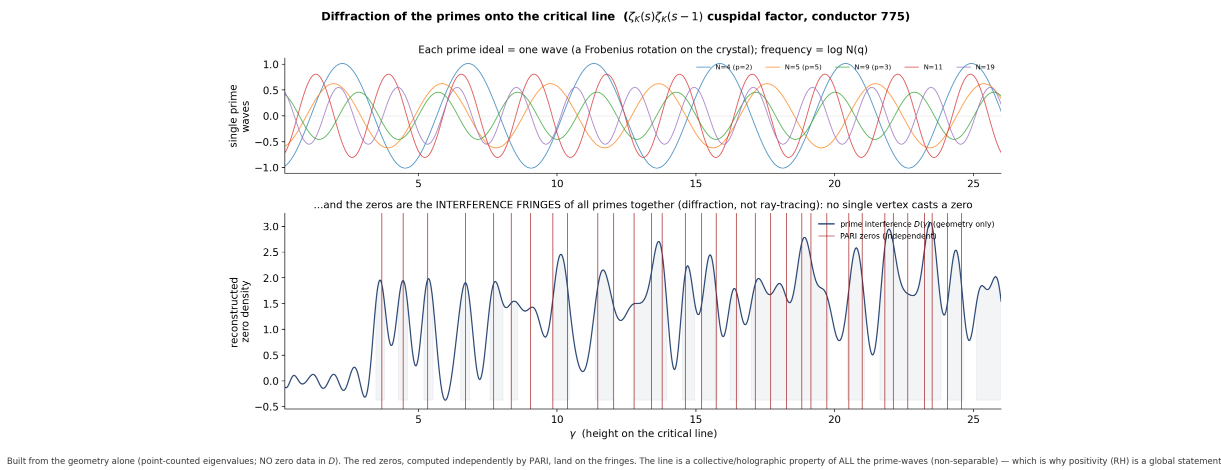

Figure 2: Each prime ideal is a wave (top); their geometry-only interference $D(\gamma)$ has fringes where the independently-computed zeros (red) sit (bottom). The zeros are a *collective* feature of all the prime-waves: each wave has global support in γ, so no single prime determines a zero.

from the geometry alone — **no zero data enters** D — the interference has fringes exactly where the 33 PARI zeros lie (Fig. 2). This is the explicit formula visualized, not a new identity.

Remark 3.1 ("holographic" — a generic Fourier fact, stated honestly). Because every prime-wave has global support in γ, any subset of primes Fourier-reconstructs the *whole* zero-set at degraded resolution; the peaks broaden as the subset shrinks (Fig. 3). This is the generic incomplete-Fourier-data property of *any* band-limited dual pair, not a special property of primes, and whether a given peak is "located" at a fixed threshold is itself resolution-dependent (e.g. a small random subset can locate all 33 peaks at a loose threshold while a larger one misses one at the same threshold). We include it only as an intuition for the global coupling, and claim nothing beyond the Fourier-duality fact. (The word "hologram" is used in this exact information-theoretic sense; no physical or optical claim is made.)

4 Calibrated Weil margins, reported honestly

Calibration. The weight-2 archimedean normalization was fixed against the elliptic-curve L of conductor 11 (degree 2, weight 2, no pole — the structure of one icosian factor): with Γ-shifts $\{\frac{1}{2}, \frac{3}{2}\}$ per place and the universal prime-side factor, the explicit formula closes to max $|\text{residual}| < 10^{-5}$ across widths. (This caught and fixed two normalization errors a ζ-only check structurally cannot detect.)

The geometric Weil form, and why its positivity is not yet evidence. Building Q_A from the Γ-factor and the geometric $a_{\mathfrak{q}}$ *only* (no zero data), the form is positive semidefinite on every tested truncation (m Hermite test functions, prime support to 3000), with an explicit Cholesky sum-of-squares at each; and it matches the spectral Gram matrix of the 33 found zeros to relative difference $< 10^{-3}$, confirming it is the genuine Weil form. *But we draw no asymptotic conclusion*: the smallest eigenvalue is small and *non-monotone* in the basis size ($0.046, 0.014, 0.0045, 0.0016$ for $m = 3, \ldots, 6$, rebounding to ≈ 0.05–0.08 at $m = 8$, reflecting basis ill-conditioning), and a finite positivity margin that does not stabilize is consistent with either limiting outcome. The

Figure 3: Reconstructions from decreasing fractions of the primes: the whole zero-set persists with broadening peaks — the generic global-support property of a Fourier dual pair (Remark 3.1), not a claim special to primes.

finite PSD is therefore *not* evidence of asymptotic positivity; it is, by the explicit-formula identity, essentially a restatement that the computed zeros are real.

The forcing frontier. What would force the positivity? Per-prime Ramanujan bounds (verified on all 664 $a_{\mathfrak{q}}$) cannot: if they could, Ramanujan would imply RH(L), which it does not. The qualitative Sato–Tate limit is a theorem [5], so it too forces nothing. The open, RH-adjacent content is the *rate* of equidistribution — the discrepancy D_n and its decay, tied to the analytic behaviour of the symmetric-power L-functions (effective Sato–Tate [6]); $\sqrt{n}\,D_n$ bounded is the square-root-cancellation regime, for which Fig. 1 (right) is evidence consistent with — not probative of — over the computed range.

Conjecture 4.1 (geometric forcing — the programme's frontier). *The closure geometry of $\mathcal{I}$ forces the square-root-cancellation rate of equidistribution of its Brandt spectrum; equivalently, the geometric Weil form is uniformly positive. This is* open *and, by [1], equivalent in conclusion to* RH(L) *— a restatement of the analytic frontier in geometric terms, not a route around it.*

5 RH(L) as the all-places analogue of a finite dichotomy

The localization rests on a classical fact, honestly labelled.

Proposition 5.1 (trace-form signature, classical). *Let K be totally real, $B(x,y) = \mathrm{Tr}_{K/\mathbb{Q}}(xy)$ its (positive-definite) trace form, and $\alpha \in \mathcal{O}_K$. Then multiplication by α is B-self-adjoint, and is B-positive — equivalently the twisted form $(x,y) \mapsto \mathrm{Tr}_{K/\mathbb{Q}}(\alpha xy)$ is positive definite —* iff *α is totally positive (positive at every archimedean place). This is the trace-form signature theorem; finitely it is the spectral theorem, since the eigenvalues of multiplication by α are exactly its conjugates.*

We verify Proposition 5.1 exactly on 7 elements across $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Q}(\sqrt{2})$, and a gate rejects the *same* operator paired with a non-invariant form ($B = I$), confirming that the right form is load-bearing. (Root-lattice Gram matrices and the closure operators C_φ of the 600- and 24-cells satisfy the *distinct* predicate of positive-definiteness, also classical; we do not conflate the two predicates, and the registry serves only to exhibit both finite faces.)

We read $\mathrm{RH}(L)$ as the infinite-dimensional *analogue* of this picture, and we state the analogy as an analogy, not a classification. Two cautions are essential. First, the trace form on $\mathcal{O}_K$ and the Weil form Q_A are *different* objects: $\mathrm{RH}(L)$ is not literally an instance of Proposition 5.1. Second — and this is the precise sense — $\mathrm{RH}(L)$ is the positivity of the *single global* Weil form $Q_A = \mathrm{ARCH} - \sum_{\mathfrak{q}} \mathrm{PRIME}_{\mathfrak{q}}$, *not* positivity place-by-place; indeed the individual prime terms are not separately positive (§4). What the two share is only the shape "the object is positive under its invariant form," with the obstruction living at the places. The honest content is therefore a placement, not a reduction:

> The finite members of this family — multiplication operators under the trace form — are governed by the classical signature theorem (Prop. 5.1) and are decidable. The icosian cuspidal L is the infinite-dimensional member, where "positive under the invariant form" is positivity of one global Weil form aggregating all places, and there it is exactly $\mathrm{RH}(L) = \mathrm{GRH}$ for one cuspidal L. The geometric quantity that would settle it is the *rate* of equidistribution of the object's spectrum (Conj. 4.1), for which §2–§4 give consistent — not probative — evidence.

6 What is not claimed

No proof of the Riemann Hypothesis or any GRH. $\mathrm{RH}(L)$ is GRH for one cuspidal L-function, not classical RH; classical ζ enters the object only through its Eisenstein face, while Q_A lives on the cuspidal face. The signature statement (Prop. 5.1) is classical; the contribution is the *computed zeros* and the two-algorithm provenance, the prime-wave realization, and the all-places localization. The finite Weil margins are explicitly *not* evidence of asymptotic positivity (§4). The "hologram" is the information-theoretic Fourier-duality fact of Remark 3.1; no physical, optical, or cosmological claim is made or used.

References

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